Disaster-Response Wi-Fi Beacon Locator

AI + IoT + Edge Computing + Drone-Assisted Search & Rescue

Team: Geo Nexus (SVIST)

Project: HillGuard App & Wi-Fi Locator Ecosystem



Executive Summary

During natural disasters such as earthquakes, floods, and hurricanes, traditional communication networks frequently fail, leaving survivors isolated and rescue teams operating blindly. The 'Disaster-Response Wi-Fi Beacon Locator,' developed by Team Geo Nexus (SVIST) under the project name HillGuard, presents a robust, scalable, and highly advanced technological solution to this critical challenge. Operating on the core philosophy that 'when networks fail, every signal can guide a search,' our solution leverages a powerful combination of Edge Computing, IoT, Artificial Intelligence (AI), and Drone-Assisted search mechanisms to detect, prioritize, and locate survivors using the passive Wi-Fi signals emitted by their personal devices.

By utilizing the state-of-the-art Arduino UNO Q as the hardware backbone, the system employs edge processing to execute Machine Learning (ML) algorithms (such as Random Forest) locally, ensuring uninterrupted operation even in offline-first scenarios. The integration of aerial drones equipped with thermal and RGB cameras further supplements ground-level IoT node data, enabling human and animal rescue classification. This comprehensive report details the system architecture, workflow, technical approach, validation metrics, and business model of our proposed solution for the Smart India Hackathon.

1. Introduction and Problem Statement

In the critical 'golden hours' following a catastrophic event, rapid localization of survivors is paramount to minimizing casualties. Existing search and rescue (SAR) operations heavily rely on visual inspections, canine units, and operational cellular networks. However, large-scale disasters inevitably destroy cellular infrastructure, rendering traditional tracking methods obsolete.

The Disaster-Response Wi-Fi Beacon Locator addresses this infrastructure vacuum by capitalizing on a ubiquitous resource: the Wi-Fi capability of modern smartphones and wearables. Even without an active internet connection, smart devices continuously emit probe requests looking for known networks. By capturing these permitted wireless transmissions and analyzing their Received Signal Strength Indicator (RSSI) values across multiple decentralized nodes, our system can triangulate and estimate the likely zones of survivors. The HillGuard application serves as the command center, guiding rescue personnel and autonomous drones through risk-aware, priority-ranked search routes.

2. System Architecture & Solution Workflow

Our solution workflow is defined by a rigorous four-step pipeline: Detect (SENSE), Analyze (FUSE), Locate (PRIORITIZE), and Save (RESPOND). This pipeline bridges the gap between raw wireless signals and actionable rescue missions.

2.1 SENSE: Multi-Source Data Collection

The foundational layer relies on a distributed network of IoT sensors and scanning nodes deployed at safe, known locations. These nodes continuously scan for permitted Wi-Fi transmissions and capture RSSI data. In addition to Wi-Fi tracking, environmental sensors monitor crucial parameters such as temperature, humidity, water level, and motion. Aerial observations from drones, capturing both thermal and RGB imagery, supplement the ground nodes to create a robust, multi-modal sensory network.

2.2 FUSE: Data Integration at the Edge

Raw data collected by the scanning nodes is virtually useless without synchronization and context. The 'Fuse' stage is executed directly at the edge to build a unified view of the disaster zone. Signal measurements (RSSI, device information, timestamps) are combined with environmental sensor readings. Crucially, GPS coordinates from both stationary ground nodes and moving drones are merged with aerial observations. This process eliminates data silos, feeding a holistic data structure to the decision engine.

2.3 PRIORITIZE: ML-Based Search Zone Estimation

The fused data pipeline drives a Machine Learning (ML) inference engine that performs Search Zone Estimation. Using a Random Forest classifier, the system analyzes the multifaceted data to estimate likely search zones along with confidence scores. Instead of providing a single, error-prone pinpoint coordinate, the model generates a spatial probability distribution—ranking areas by urgency, risk level, and the likelihood of human presence. This allows for uncertainty representation on the map, dynamically adjusting as new signals are acquired.

2.4 RESPOND: Guide & Coordinate Rescue Operations

The final phase translates analytics into physical rescue operations. The HillGuard Command Dashboard displays high, medium, and low priority zones over a live interactive map (e.g., integrating OpenStreetMap data). It dispatches autonomous or operator-guided drones for further inspection and guides human rescue teams with dynamic route suggestions that avoid known hazards.

3. Technical Approach & Hardware Backbone

3.1 The Arduino UNO Q Advantage

The hardware backbone of the HillGuard system is the Arduino UNO Q, chosen for its unparalleled status as a high-performance, all-in-one development board. The UNO Q represents a paradigm shift for rapid, industrial-grade prototyping, making it ideal for the COMET Hackathon.

- Key features utilized in our architecture include:
- Dual Processor Setup: Incorporating the STM32U585 Microcontroller for real-time sensor data acquisition, peripheral control, and GPIO/IO tasks.

- Qualcomm Dragonwing QRB2210 Processor: A robust Linux-based processor dedicated to running the Python application stack, ML inference, data fusion, local storage handling, and the backend dashboard.
- Advanced Communications: Built-in dual-band Wi-Fi 5 and Bluetooth 5.1. It also features a USB interface ready for optional 5G modules and future 6G research, ensuring the solution is future-proof.
- Edge ML Support: Native capacity for running sensor fusion and Machine Learning tasks completely offline.

3.2 Data Processing & Feature Engineering

Data streaming from the Wi-Fi scanners and IoT nodes undergoes a stringent cleaning and validation phase. Timestamps and geospatial coordinates are synchronized across all devices. We extract several critical features for our ML models, including:
- RSSI fluctuation patterns over time.
- Signal observation frequency (repeated signals indicate stationary targets or trapped individuals).
- Sensor and device status metadata.
- Combined GPS and drone visual scene cues.

The system explicitly accounts for signal and location uncertainty, a crucial aspect when dealing with RSSI multipath fading in debris-heavy environments.

3.3 7-Stage Pipeline & Machine Learning Models

The intelligent localization is powered by a 7-Stage Pipeline spanning from the IoT Sensing Layer to Rescue Actions. For the core intelligence, we utilize an ensemble learning approach. A Random Forest model serves as our primary baseline due to its robust explainability, low tendency to overfit on noisy data, and efficient edge computation. It acts to classify candidate zones as high or low priority based on RSSI counts, locations, and environment contexts. Advanced extensions include a Regression Model for distance estimation from calibrated signal features, Computer Vision modules for parsing drone RGB/Thermal imagery, and Sensor Anomaly Detection to flag unusual environmental changes (like sudden water level spikes or motion).

4. Drone-Assisted Search & Rescue Ecosystem

Drones serve as a massive force multiplier in the HillGuard ecosystem. Deployed to inspect inaccessible or hazardous terrains, they act as both mobile Wi-Fi scanners and aerial visual posts.

4.1 Mission Profiles and Workflows

Drone missions follow a structured 'Survey -> Observe -> Report -> Rescuers Verify' workflow. During the Survey phase, the drone autonomously traverses operator-approved search routes, targeting zones flagged by ground-based Wi-Fi scanners. As it observes, it captures RGB imagery and thermal data.

4.2 Thermal + Visual Detection (Human vs. Animal)

By fusing thermal signatures with visual data, the command dashboard can flag possible targets. A critical innovation in our pipeline is the 'Human and Animal Rescue Classification.' Using onboard edge AI, the system differentiates between human and animal heat signatures, dispatching appropriate rescue teams (e.g., medical personnel for humans, specialized animal rescue for pets/livestock). Ambiguous cases are routed for manual operator review, drastically reducing false positives.

4.3 Drone-Mounted Wi-Fi Scanning

The drones themselves carry lightweight scanning payloads. Combining drone position data with observed wireless transmissions allows for 3D localization of signals. It is important to note that thermal signatures and Wi-Fi signals are indicators, not definitive proof of life; human verification remains the final step in the chain.

5. Smart Wireless Localization & Key Operational Features

The 'HillGuard' platform implements 11 distinct operational stages to ensure faster response and safer lives:

1. Deploy Multiple Scanning Receivers: Placing nodes at safe, known locations to create a baseline network.
2. Record Measurements: Logging time, receiver locations, and signal strength for observable transmissions.
3. Compare Measurements: Analyzing data from different receivers or moving receivers (drones) to triangulate sources.
4. Estimate a Likely Region: Displaying uncertainty areas rather than exact, potentially misleading coordinates.
5. Send Rescue Team or Drone: Acting upon the mapped data.
6. Offline-First Disaster Mode: A critical feature ensuring continuous scanning and local logging when the internet is entirely unavailable, syncing only when the connection returns.
7. Search-Priority Heatmap: A visual dashboard element indicating priority zones based on repeated Wi-Fi evidence, drone observations, and terrain context.
8. Power-Aware Operation: Tracking battery and system health for each node. The system autonomously switches to low-power scanning if energy drops, alerting the team before a node dies.
9. Privacy-Preserving Detection: Maintaining ethical standards by ONLY storing signal strength, timestamp, and approximate location. The system explicitly DOES NOT collect personal device data, identity information, or payload data. Short retention periods are strictly enforced.
10. Human and Animal Rescue Classification: Streamlining the deployment of specialized resources.
11. Risk-Aware Search Routing: Ranking safe, accessible search routes using known hazards and mapped blocked paths. System requires human approval before dispatching teams to dangerous areas.

6. Prototype Validation Metrics & Setup

To prove the efficacy of the HillGuard system, rigorous testing protocols have been established in controlled test areas featuring multiple distances, real devices, background noise devices, and physical obstructions to mimic disaster debris.

- Target Validation Metrics include:
- Detection Rate: >= 90% (successfully identifying present devices).
- False-Alarm Rate: <= 10%.
- Localization Error: <= 50 meters (sufficient to guide a drone or dog team to a close-proximity search).
- Time to Identify a Test Device: <= 2 minutes.
- Offline Processing Capacity: >= 70% of data processed entirely at the edge without cloud reliance.

7. Deployment Scenarios & Business Model

The modularity of the HillGuard system makes it highly scalable and adaptable to diverse environments, including mountainous regions, dense forests, landslide zones, and urban flood plains. Its integration with existing disaster response systems ensures immediate operational viability.

Our sustainable business model revolves around three pillars:
1. Product Sales: Supplying hardware kits containing Arduino UNO Q scanners, sensors, and software licenses.
2. Subscription Services: Offering cloud sync capabilities, advanced map analytics, and automated alert systems.
3. Strategic Partnerships: Engaging with Government & Disaster Agencies (like NDMA and SDRF) for public safety deployments, as well as NGOs and enterprise research sectors for field development.

8. Limitations & Future Extensions

While highly effective, the system acknowledges certain limitations: Wi-Fi signals can be weak or inconsistent under dense rubble, extreme weather or terrain can affect drone flights, and localization is a regional estimate rather than a guaranteed exact location. It heavily relies on initial calibration and training data.

Future Practical Extensions:
- Emergency Supply Relay: Utilizing heavy-lift drones to deliver lightweight survival kits, radios, or markers to verified survivors.
- 6G Integration: Expanding the hardware payload to research and leverage 6G communication standards for even more precise spatial localization and micro-Doppler radar sensing for vital sign monitoring.

9. Conclusion

Team Geo Nexus has designed the Disaster-Response Wi-Fi Beacon Locator to transform reactive disaster management into a proactive, data-driven science. By bridging AI, IoT, Edge Computing, and Drone technologies on the robust Arduino UNO Q platform, we provide a lifeline where traditional infrastructure fails. Mapping today to save tomorrow, this project stands as a testament to the life-saving potential of innovative wireless technology in the COMET Hackathon and beyond.

10. References

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